UNIVERSIDAD AUTÓNOMA DE NUEVO LEÓN

FACULTAD DE CIENCIAS FÍSICO MATEMÁTICAS

DOCTORADO EN INGENIERÍA FÍSICA

Geomagnetic Activity and Reported Aurora Sightings

2024 annual analysis • NASA DONKI support • October 2026 case study

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Why study geomagnetic activity and auroras?

PROJECT INTRODUCTION

Space weather links disturbances in Earth's magnetic environment with visible auroral activity.

We use reproducible Python processing and interactive Power BI analysis to investigate whether stronger geomagnetic activity accompanies more reported aurora sightings.

Main period: January–December 2024

THREE COMPLEMENTARY SOURCES

GFZ

Planetary Kp index

Aurorasaurus

Citizen-reported sightings

NASA DONKI

Geomagnetic storm events

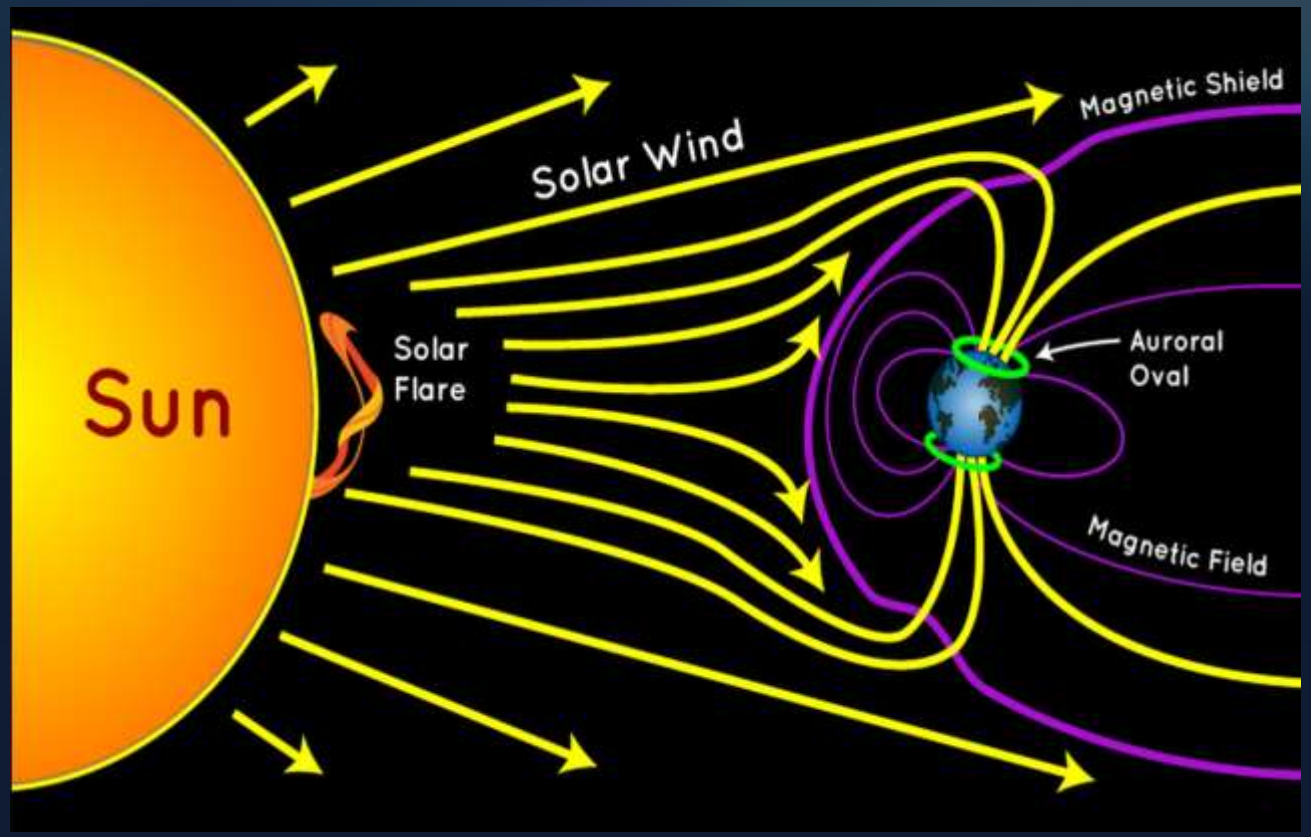
Earth’s geomagnetic field

GEOPHYSICAL BACKGROUND

Earth’s magnetic field is generated mainly by motion of conductive fluid in the outer core.

The magnetosphere interacts with the solar wind. During disturbances, energy and particles can be transferred into the near-Earth environment.

Kp is a planetary, three-hour index of geomagnetic disturbance (0–9).



How Auroras form ?

01

Solar wind

Charged particles and magnetic energy reach Earth's magnetosphere.

→

02

Magnetic coupling

Energy enters the magnetosphere and drives particle acceleration.

→

03

Atmospheric excitation

Energetic electrons collide with oxygen and nitrogen.

→

04

Visible light

Atoms and molecules emit light, forming auroral displays.

Important: a high Kp increases auroral potential but does not guarantee a sighting at a given location.

Research question, data and design

SCIENTIFIC QUESTION

Are days with higher geomagnetic Kp associated with more reported aurora sightings?

GFZ KP

2024 · 2,928 intervals

Three-hour activity measurements

AURORASAURUS

2024 · 11,465 reports

Citizen-reported observation records

NASA DONKI

2024 · 20 events

Independent storm event inventory

Daily matching for correlation; NASA storm-start dates for an independent group comparison.

Scientific computing workflow • Python to Power BI

01

APIs & archive

GFZ / NASA / Zenodo

→

02

Python

requests + pandas

→

03

Validation

UTC, gaps, duplicates

→

04

Files

JSON / CSV / Parquet

→

05

Power BI

date model + DAX

GFZ is public; NASA DONKI uses an API key stored in an environment variable.

2024 annual findings ▪ What the data shows

KP OBSERVATIONS
2,928
366 days × 8 intervals

AURORA REPORTS
11,465
Citizen observations

MEAN KP
1.92
Planetary Kp index

STORM INTERVALS
97
Three-hour intervals, Kp ≥ 5

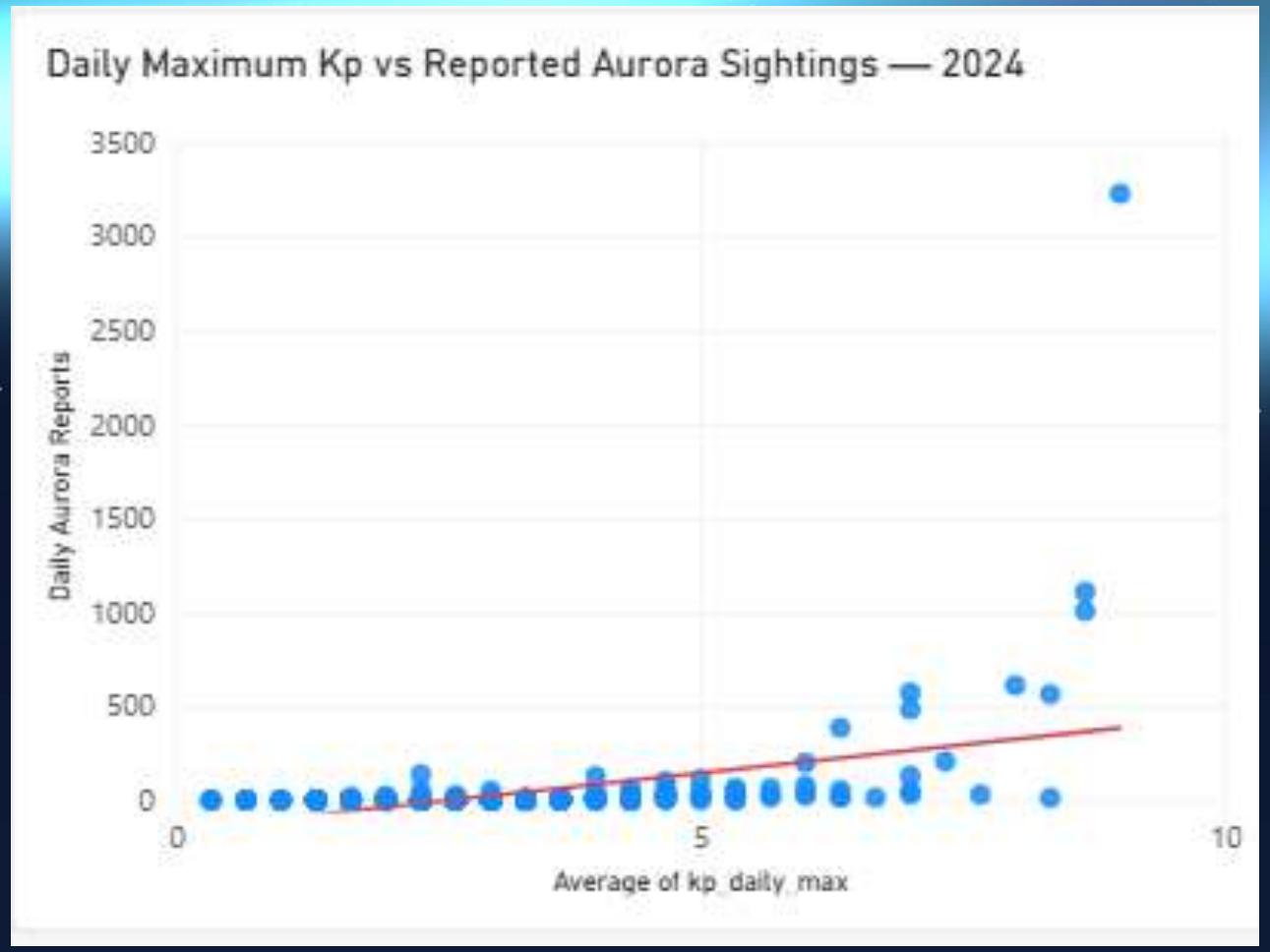


MONTHLY PATTERN

Aurora reports peak in May and rise again in October 2024.

The monthly chart is descriptive. The daily scatterplot provides the stronger association test.

Main finding • Higher Kp, more reported auroras



DAILY CORRELATION

+0.4498

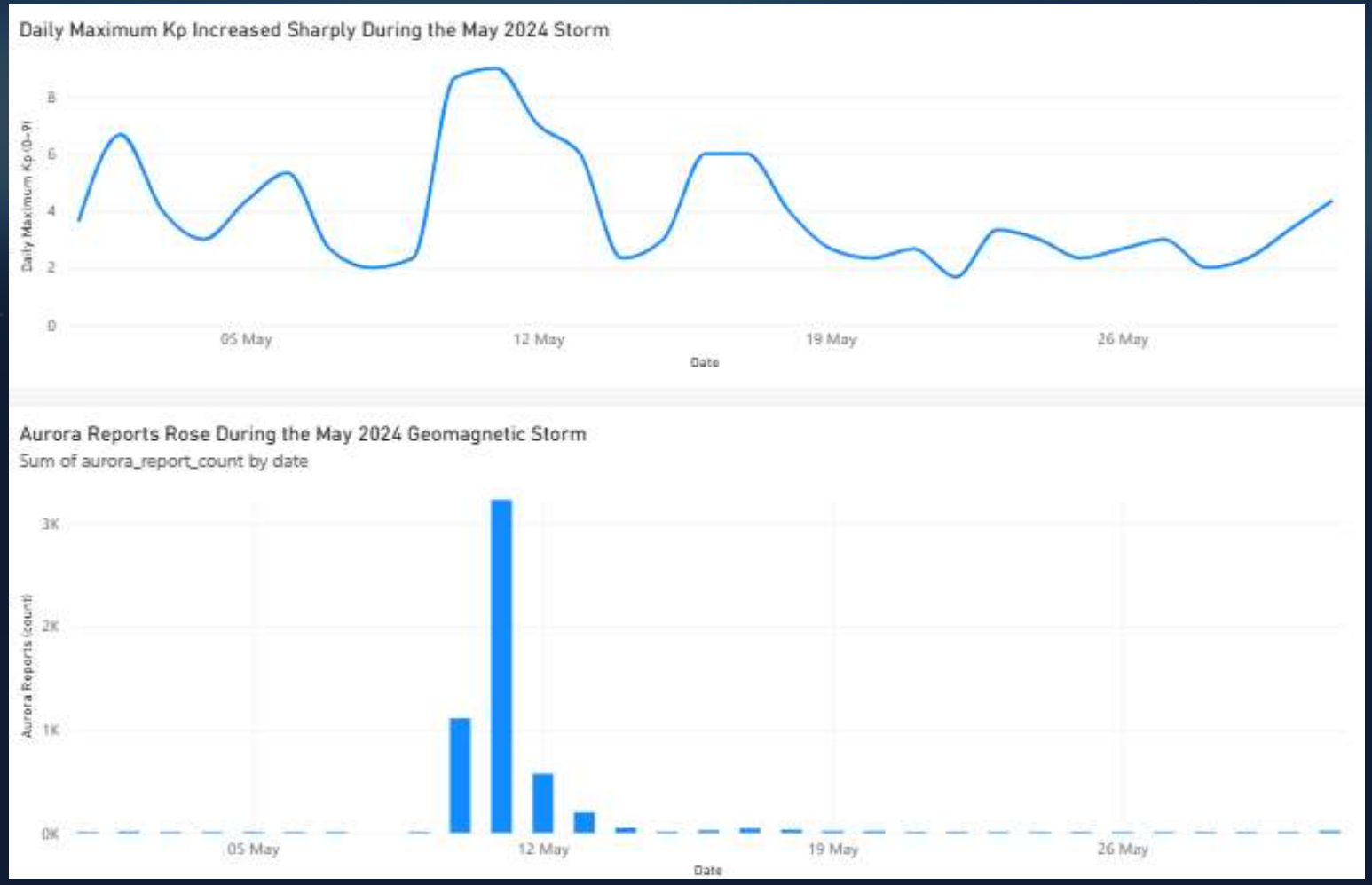
Pearson r

+0.6181

Spearman ρ

Positive association; not proof of causation.

May 2024 case study ▪ A major geomagnetic storm



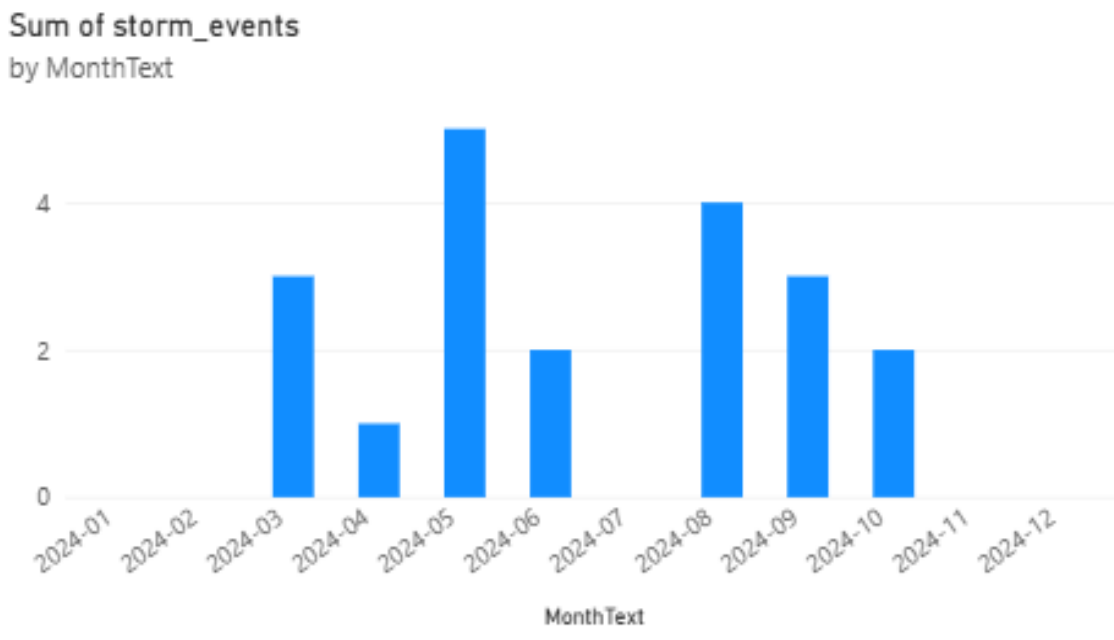
EVENT-LEVEL EVIDENCE

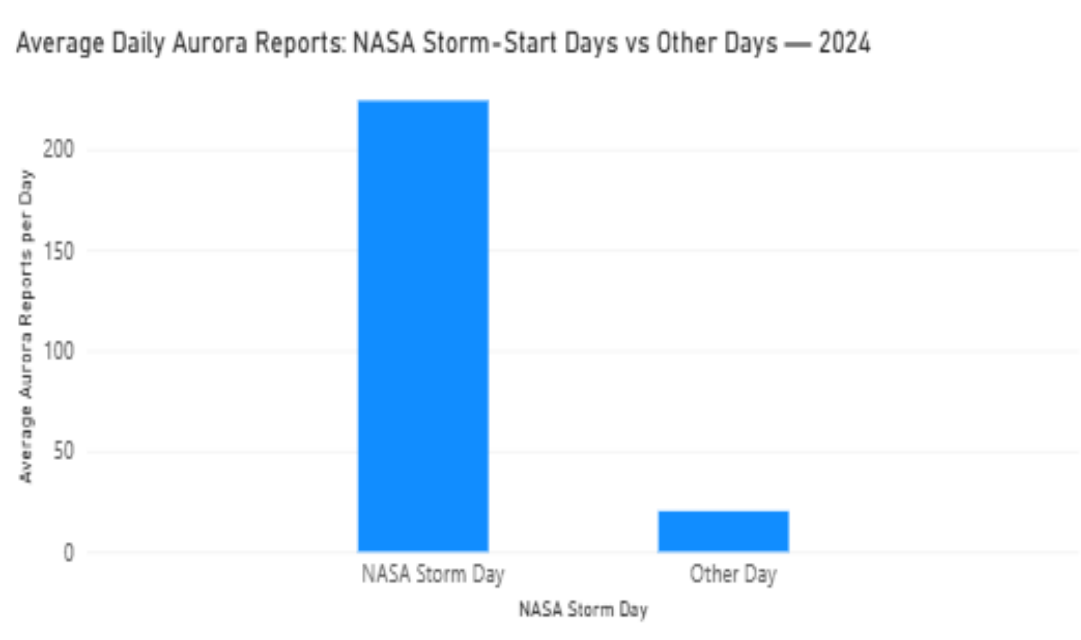
10–11 May 2024

A pronounced Kp rise coincides with a surge in aurora reports.

Separate graphs show daily maximum Kp and reported aurora observations during May 2024.

NASA DONKI • Independent storm-event support





NASA STORM EVENTS

20

2024 DONKI records

MONTHS WITH EVENTS

7

Recorded in 7 months

MAX REPORTED KP

9.0

Highest DONKI Kp

NASA storm-start days show higher average aurora report counts; event counts ≠ three-hour storm intervals.

REPRODUCIBILITY ACROSS PERIODS

The same Python → Power BI approach was first applied to October 2026.

3-HOUR KP VALUES	MEAN KP	MAXIMUM KP	STORM-LEVEL INTERVALS
248	1.93	7.333	7
October 1–31, 2026	October 1–31, 2026	10 October 2026	October 1–31, 2026

October 10 was the most disturbed day in the analyzed month. This is a separate study period, not part of the 2024 correlation.

Conclusions and reproducibility

MAIN TAKEAWAYS

01 2024 daily maximum Kp and aurora report counts show a positive association.

02 The May 2024 disturbance illustrates a strong event-level rise in reporting.

03 NASA DONKI independently documents storm-event timing; October 2026 shows workflow reuse.

LIMITATIONS

Citizen reporting depends on darkness, weather, geographic coverage and observer participation.

REPRODUCE

Python scripts + source data + Power BI + documentation

github.com/mohamedayas16/Semester_2_CS_2026